

Dynamics of the Stable Boundary Layer: A Geometric Singular Perturbation Framework for Fold Characterization and Regime Transitions

Part 1: Mathematical Foundations and Observational Resolution

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Abstract

Observed critical Richardson thresholds vary widely across field campaigns (e.g., CASES-99 and SHEBA), yielding the long-standing Richardson threshold paradox. We show that this variability is not evidence of incompatible physics, but rather the projection of a multidimensional folded critical manifold onto scalar diagnostics. To formalize this mechanism, we construct a multiscale fast-slow model of the stable boundary layer using Geometric Singular Perturbation Theory (GSPT) on a desingularized coordinate chart. The framework yields a sequence of structural results: existence and smoothness of the critical manifold, fold characterization through fast-Jacobian degeneracy, and a constant-rank projection theorem linking invariant state-space geometry to campaign-dependent observational thresholds. Within this geometry, we define an analytical sensible heat-flux turning-capacity limiter, $H_{\max}$, and a pathway toward dynamic threshold parameterization. Part 1 establishes the mathematical foundations and observational resolution needed for closure development and data-assimilative validation in Parts 2–3.

1 Introduction

The nocturnal Stable Boundary Layer (SBL) remains one of the most persistent challenges in geophysical fluid dynamics due to the highly non-linear, non-equilibrium coupling between turbulent dissipation, surface energy budgets, and mesoscale forcing (Nieuwstadt, 1984; Mahrt, 2014; van de Wiel et al., 2017). A central unresolved problem in boundary-layer meteorology is the so-called “Richardson Number Universality Crisis”: the observation that the critical Bulk Richardson number (Ri_c), traditionally expected to be a quasi-universal constant near 0.25, varies wildly across different environmental regimes (Grachev et al., 2013; Monahan et al., 2015). While mid-latitude field campaigns such as CASES-99 report collapse thresholds near the classical limit ($Ri \approx 0.2$ – 0.4), polar campaigns like SHEBA record active turbulence persisting at stability ratios exceeding unity ($Ri > 1.2$) (Poulos et al., 2002; Grachev et al., 2005).

1.1 The Paradox of Stability Thresholds

For decades, the search for a “better” stability function has assumed that the variability in Ri_c stems from inconsistent physics or measurement noise. However, traditional parameterizations—such as Monin–Obukhov Similarity Theory (MOST) or Mellor–Yamada K -theory—rely on an instantaneous local equilibrium assumption, $e = e(S, N^2)$, that erases the system’s underlying temporal history and structural hysteresis (Mellor and Yamada, 1982). This diagnostic approach forces a smooth, single-valued response that cannot account for the abrupt, “brittle” transitions between weakly stable and very stable regimes.

1.2 Thesis: Richardson Thresholds as Projections, Not Invariants

In this paper, we propose that the Richardson threshold paradox is not evidence of incompatible physics, but rather a deterministic consequence of dimensionality reduction. We frame the SBL as a singularly perturbed dynamical system governed by Geometric Singular Perturbation Theory (GSPT) (Fenichel, 1979; Kuehn, 2015). Within this framework, our central thesis is that *Richardson thresholds are low-dimensional coordinate projections of a higher-dimensional folded equilibrium manifold*.

By exploiting the natural separation of timescales—where fast turbulence adjustment ($\tau_e \sim 10^1$ – 10^2 s) responds to slow surface thermal and geostrophic evolution ($\tau_{\text{slow}} \sim 10^3$ – 10^4 s)—we demonstrate that the “critical” threshold is not a static point, but a state-dependent fold locus ($\mathcal{C}_{\text{fold}}$) that deforms dynamically in response to surface energy balance (van de Wiel et al., 2012).

1.3 The “Fold Illusion” and Emergent Coupling

A key conceptual contribution of this work is the distinction between isolated boundary-driven decay and true fold catastrophes. We prove that while an isolated atmospheric subsystem exhibits only a monotonic boundary-crossing at the laminar floor ($e = 0$), a genuine S-shaped fold only emerges through the non-linear coupling of the atmosphere to the Surface Energy Balance (SEB). We term the historical conflation of these two mechanisms the “Fold Illusion,” and show that the geometric resolution of this illusion is necessary to model intermittent bursting and nocturnal relaxation oscillations accurately.

1.4 Objectives and Scope of Part 1

As the first in a three-part series, this manuscript establishes the mathematical foundations and observational resolution of the GSPT-SBL framework. We define a hierarchy of models—from 2D analytical baselines to 5D thermodynamic systems—and provide a sequence of structural theorems:

- **Theorem 1 (Existence and Smoothness):** Establishing the critical manifold ($\mathcal{S}_0$) as a smooth embedded submanifold of the state space.
- **Theorem 2 (Fold Characterization):** Defining the geometric boundary where normal hyperbolicity is lost, triggering turbulence collapse.
- **Theorem 3 (Projection Theorem):** Formalizing the mapping from invariant state-space geometry to campaign-specific scalar diagnostics (Ri_{fold}).

Part 1 provides the “geometric skeleton” required for the data-driven discovery and prognostic parameterizations developed in Parts 2 and 3.

- 1.5 The Crisis of Stability Thresholds in Boundary-Layer Meteorology
- 1.6 The Closure Trilemma in Atmospheric Modeling
- 1.7 Objectives and Scope of Part 1

2 Multi-Scale Governing Equations and Chart Regularization

This section defines the smooth fast–slow architecture for the stable boundary layer (SBL) model. We introduce the regularized state space, state the standing assumptions used in the geometric analysis, derive the desingularized fast chart, define the observation map, and introduce the critical manifold and fold locus.

2.1 System Definition and Standing Assumptions

We work with a multiscale autonomous system written in the desingularized fast time variable τ on an open chart domain $\Omega_0 \subset \mathbb{R}^5$:

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{F}(\mathbf{x}; \boldsymbol{\mu}, \epsilon_1, \epsilon_2) \equiv \begin{pmatrix} F_{\text{fast}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 F_{\text{slow}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 \epsilon_2 F_{\text{superslow}}(\mathbf{x}; \boldsymbol{\mu}) \end{pmatrix}. \quad (1)$$

The full state vector is $\mathbf{x} = (\tilde{e}, \hat{q}_\theta, S, T_s, T_g)^T \in \Omega_0 \subset \mathbb{R} \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+$. The physical system comprises a five-dimensional state space ($n = 5$) partitioned into three distinct timescales:

- **Fast Subsystem** ($n_{\text{fast}} = 2$): Desingularized TKE coordinate $\tilde{e} \in \mathbb{R}$ and scaled turbulent heat flux coefficient $\hat{q}_\theta \in \mathbb{R}$ (where physical flux $q_\theta = \tilde{e}\hat{q}_\theta$).
- **Slow Subsystem** ($n_{\text{slow}} = 2$): Bulk vertical wind shear $S \in \mathbb{R}_+$ and surface skin temperature $T_s \in \mathbb{R}_+$.
- **Super-Slow Subsystem** ($n_{\text{superslow}} = 1$): Deep-soil temperature $T_g \in \mathbb{R}_+$.

Standing Assumptions. Throughout this manuscript we assume:

- (A1) **Domain regularity.** The chart domain $\Omega_0 \subset \mathbb{R}^5$ is open and connected, extending across the laminar boundary $\tilde{e} = 0$, and the vector field $\mathbf{F}$ is of class C^∞ on Ω_0 .
- (A2) **Parameter domain.** The parameter vector $\boldsymbol{\mu} \in \mathcal{P}$ belongs to an open bounded set of physically admissible coefficients containing the turbulence closure, radiative, and thermodynamic constants.
- (A3) **Flux scaling and positive invariance.** The turbulent heat flux satisfies $q_\theta = \tilde{e}\hat{q}_\theta$, so that turbulent transport vanishes at the laminar limit $\tilde{e} = 0$. Consequently, the physical domain

$$\Omega_{\text{phys}} = \{\mathbf{x} \in \Omega_0 \mid \tilde{e} > 0, S > 0, T_s > 0, T_g > 0\} \quad (2)$$

and its closure $\bar{\Omega}_{\text{phys}}$ are positively invariant under the flow (van de Wiel et al., 2017).

- (A4) **Local well-posedness.** For each initial state $\mathbf{x}(0) \in \Omega_0$, there exists a unique local trajectory $\mathbf{x}(\tau) \in C^\infty((-\delta_t, \delta_t), \Omega_0)$.
- (A5) **Timescale separation.** The singular perturbation parameters satisfy

$$0 < \epsilon_1 \ll 1, \quad 0 < \epsilon_2 \ll 1, \quad (3)$$

separating fast turbulent adjustment ($O(1)$ time τ), slow boundary-layer shear and thermal evolution ($O(\epsilon_1^{-1})$ time), and super-slow deep-soil heat conduction ($O((\epsilon_1 \epsilon_2)^{-1})$ time) (Kuehn, 2015; Kristiansen, 2024).

2.2 Chart Regularization and Desingularization

In the physical TKE coordinate e , the fast equations contain non-smooth fractional exponents ($e^{1/2}$ shear production and $e^{3/2}$ Kolmogorov dissipation) that cause non-differentiability as $e \rightarrow 0^+$. This poses severe challenges for geometric analysis and numerical continuation (De Maesschalck and Desroches, 2013; Steinebach, 2023).

To restore C^∞ smoothness and support automatic differentiation via tools like `ForwardDiff.jl` (Rackauckas and Nie, 2017), we introduce the regularized chart map:

$$\phi_\delta : e \mapsto \tilde{e} = \sqrt{e + \delta}, \quad \delta > 0. \quad (4)$$

Proposition 1 (Desingularization and smooth extension). *For fixed $\delta > 0$, the transformed vector field is C^∞ smooth on the interior chart domain $\tilde{e} > \sqrt{\delta}$. The limiting desingularized chart obtained as $\delta \rightarrow 0^+$ under the positive time reparameterization*

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}}, \quad \text{i.e.,} \quad \frac{dt}{d\tau} = \epsilon_1 \tilde{e}, \quad (5)$$

extends smoothly across the laminar boundary $\tilde{e} = 0$ to yield a polynomial fast subsystem on Ω_0 . Because

$$\frac{d\tau}{dt} = \frac{1}{\epsilon_1 \tilde{e}} > 0 \quad \text{on } \Omega_{\text{phys}}, \quad (6)$$

this transformation is an orientation-preserving diffeomorphic time rescaling that preserves phase trajectories, equilibrium sets, and the signs of transverse eigenvalues along normally hyperbolic manifold branches.

Proof. For $\delta > 0$ and $e > 0$, differentiating $\tilde{e}^2 = e + \delta$ with respect to physical time t yields $2\tilde{e} \frac{d\tilde{e}}{dt} = \frac{de}{dt}$. Substituting the physical TKE evolution equation $\epsilon_1 \frac{de}{dt} = c_m \ell e^{1/2} S^2 - \frac{g}{\theta_0} \tilde{e} \hat{q}_\theta - \frac{1}{\ell} e^{3/2}$ gives:

$$\frac{d\tilde{e}}{dt} = \frac{1}{2\tilde{e}\epsilon_1} \left(c_m \ell \sqrt{\tilde{e}^2 - \delta} S^2 - \frac{g}{\theta_0} \tilde{e} \hat{q}_\theta - \frac{1}{\ell} (\tilde{e}^2 - \delta)^{3/2} \right). \quad (7)$$

For fixed $\delta > 0$, the term $\sqrt{\tilde{e}^2 - \delta}$ is C^∞ on $\tilde{e} > \sqrt{\delta}$. Applying the time transformation $\frac{d\tilde{e}}{d\tau} = \frac{dt}{d\tau} \frac{d\tilde{e}}{dt} = (\epsilon_1 \tilde{e}) \frac{d\tilde{e}}{dt}$ cancels $\epsilon_1 \tilde{e}$ in the denominator:

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} \left(c_m \ell \sqrt{\tilde{e}^2 - \delta} S^2 - \frac{g}{\theta_0} \tilde{e} \hat{q}_\theta - \frac{1}{\ell} (\tilde{e}^2 - \delta)^{3/2} \right). \quad (8)$$

Taking the limiting desingularized chart as $\delta \rightarrow 0^+$ yields:

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} c_m \ell S^2 \tilde{e} - \frac{g}{2\theta_0} \tilde{e} \hat{q}_\theta - \frac{1}{2\ell} \tilde{e}^3 = \tilde{e} \left(\frac{1}{2} c_m \ell S^2 - \frac{g}{2\theta_0} \hat{q}_\theta - \frac{1}{2\ell} \tilde{e}^2 \right), \quad (9)$$

which is C^∞ polynomial in $\tilde{e}$ on all of Ω_0 , including $\tilde{e} = 0$. Because $\frac{d\tau}{dt} = (\epsilon_1 \tilde{e})^{-1} > 0$ on Ω_{phys} , orbit orientations and normal hyperbolicity are preserved (Fenichel, 1979; Kuehn, 2015). $\square$

2.3 Subsystem Vector Field Components

Under Assumptions (A1)–(A5) and Proposition 2.1, the complete desingularized vector field $\mathbf{F}(\mathbf{x})$ is given explicitly as follows.

Fast subsystem ($n_{\text{fast}} = 2$). Governs rapid turbulent adjustment of $(\tilde{e}, \hat{q}_\theta)$:

$$\tilde{F}(\mathbf{x}) = \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}\tilde{e}\hat{q}_\theta - \frac{1}{2\ell}\tilde{e}^3, \quad (10)$$

$$\tilde{H}(\mathbf{x}) = -c_w\theta_z(T_s)\tilde{e}^3 - \frac{g}{\theta_0}c_\theta\ell\tilde{e}^2\hat{q}_\theta^2 - \frac{C_\theta}{\ell}\tilde{e}^3\hat{q}_\theta, \quad (11)$$

where $\theta_z(T_s) = \frac{\theta_{\text{top}} - \Pi_s T_s}{\Delta z}$ is the background potential temperature gradient.

Slow subsystem ($n_{\text{slow}} = 2$). Governs shear production S and skin temperature T_s :

$$F_S(\mathbf{x}) = \tilde{e} \left[\mathcal{F}_{\text{ls}} - \frac{\partial}{\partial z}(c_m\ell\tilde{e}S) \right], \quad (12)$$

$$F_{T_s}(\mathbf{x}) = \frac{\tilde{e}}{C_s} \left[R_{\text{net}}(T_s) + \rho c_p \tilde{e} \hat{q}_\theta + \frac{k_g}{d_g}(T_g - T_s) \right], \quad (13)$$

where $R_{\text{net}}(T_s) = R_{\text{sw}}^\downarrow + \epsilon_a \sigma T_a^4 - \epsilon_s \sigma T_s^4$. The regularizing prefactor $\tilde{e}$ causes slow processes to dynamically “freeze” in fast time as the system approaches the laminar floor ($\tilde{e} \rightarrow 0$).

Super-slow subsystem ($n_{\text{superslow}} = 1$). Governs soil thermal inertia:

$$F_{\text{superslow}}(\mathbf{x}) = \tilde{e} \left[\frac{k_g}{d_g^2}(T_s - T_g) \right]. \quad (14)$$

We emphasize that reduced slow dynamics are obtained by restricting the desingularized flow to the attracting Fenichel manifold $\mathcal{S}_\epsilon$, rather than by evaluating the slow vector field on the singular boundary alone.

2.4 Observation Mapping and Diagnostic Functionals

To project internal state-space dynamics onto observable boundary-layer quantities, we define the smooth observation operator

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \rightarrow \mathcal{O} \subset \mathbb{R}^3, \quad \mathbf{\Pi}_{\text{obs}}(\mathbf{x}) = \begin{pmatrix} \pi_{Ri}(\mathbf{x}) \\ \pi_H(\mathbf{x}) \\ \pi_S(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \\ \rho c_p \tilde{e} \hat{q}_\theta \\ S \end{pmatrix}. \quad (15)$$

Fast fiber annihilation. Let $T_{\mathbf{x}}\mathcal{F}_{\text{fast}} = \text{span}\{\frac{\partial}{\partial \tilde{e}}, \frac{\partial}{\partial \hat{q}_\theta}\} \subset T_{\mathbf{x}}\Omega_0$ denote the tangent space to the fast fibers. In the state-space cotangent basis $\{d\tilde{e}, d\hat{q}_\theta, dS, dT_s, dT_g\}$, the differential Jacobian of the Bulk Richardson diagnostic is:

$$D\pi_{Ri}(\mathbf{x}) = \begin{pmatrix} 0 & 0 & -\frac{2g\theta_z(T_s)}{\theta_0 S^3} & \frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial T_s} & 0 \end{pmatrix}. \quad (16)$$

Consequently, $D\mathbf{\Pi}_{\text{obs}}(\mathbf{v}) = \mathbf{0}$ for all fast vectors $\mathbf{v} \in T_{\mathbf{x}}\mathcal{F}_{\text{fast}}$. This proves that the Richardson diagnostic is aligned strictly with the slow cotangent subspace and is insensitive at first order to fast turbulent fluctuations $(\tilde{e}, \hat{q}_\theta)$.

2.5 Geometric Foundations: Critical Manifold and Fold Locus

We conclude this section by explicitly defining the invariant geometric objects that govern SBL transitions.

[Critical Manifold $\mathcal{S}_0$] Setting $\epsilon_1 = 0$ defines the critical manifold as the equilibrium set of the fast subsystem:

$$\mathcal{S}_0 = \{\mathbf{x} \in \Omega_0 \mid F_{\text{fast}}(\mathbf{x}; \boldsymbol{\mu}) = \mathbf{0}\}. \quad (17)$$

Since $n = 5$ and $n_{\text{fast}} = 2$, $\mathcal{S}_0$ is a 3-dimensional embedded submanifold of Ω_0 .

[Fold Locus $\mathcal{C}_{\text{fold}}$] The fold locus is the set of non-hyperbolic points on $\mathcal{S}_0$ where the fast Jacobian loses rank:

$$\mathcal{C}_{\text{fold}} = \{\mathbf{x} \in \mathcal{S}_0 \mid \det D_{(\tilde{e}, \hat{q}_\theta)} F_{\text{fast}}(\mathbf{x}; \boldsymbol{\mu}) = 0\}. \quad (18)$$

[Observed Threshold Image Γ_{fold}] The campaign-specific critical Richardson threshold observed in field data is the image of the fold locus under the observation projection:

$$\Gamma_{\text{fold}} = \boldsymbol{\Pi}_{\text{obs}}(\mathcal{C}_{\text{fold}}) \subset \mathcal{O}. \quad (19)$$

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